POLITECNICO DI TORINO

Master's Degree in Electronics

Master's Degree Thesis

Loss analysis of a 4-level flying capacitor Totem-Pole PFC

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Chapter 1

Hello

[Hi 1, Goofy]
$\mathrm{kg\,s^{-1}}$

Figure 1.1: Hi

1.1 Extremely long name with manual linebreak which otherwise would not fit the page

1. A
2. B
3. C

Figure 1.2: HI

Figure 1.3: SVG

$$output = f_{activation}\left(\sum_{\#neurons} input_i + bias\right) \tag{1.1}$$

- A
- B
- C

Algorithm 1 Adam optimizer algorithm. All operations are element-wise, even powers. Good values for the constants are $\alpha = 0.001, \beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$. ϵ is needed to guarantee numerical stability.

procedure ADAM$(\alpha, \beta_1, \beta_2, f, \theta_0)$
  ▷ α is the stepsize
  ▷ $\beta_1, \beta_2 \in [0, 1)$ are the exponential decay rates for the moment estimates
  ▷ $f(\theta)$ is the objective function to optimize
  ▷ θ_0 is the initial vector of parameters which will be optimized
  ▷ Initialization
  $m_0 \leftarrow 0$ ▷ First moment estimate vector set to 0
  $v_0 \leftarrow 0$ ▷ Second moment estimate vector set to 0
  $t \leftarrow 0$ ▷ Timestep set to 0
  ▷ Execution
  while θ_t not converged **do**
    $t \leftarrow t + 1$ ▷ Update timestep
    ▷ Gradients are computed w.r.t the parameters to optimize
    ▷ using the value of the objective function
    ▷ at the previous timestep
    $g_t \leftarrow \nabla_\theta f(\theta_{t-1})$
    ▷ Update of first-moment and second-moment estimates using
    ▷ previous value and new gradients, biased
    $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$
    $v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$
    ▷ Bias-correction of estimates
    $\hat{m}_t \leftarrow \dfrac{m_t}{1 - \beta_1^t}$
    $\hat{v}_t \leftarrow \dfrac{v_t}{1 - \beta_2^t}$
    $\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \dfrac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$ ▷ Update parameters
  end while
  return θ_t ▷ Optimized parameters are returned
end procedure

ReLU	$f(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ x & \text{for } x > 0 \end{cases}$
Softmax	$f_i(\vec{x}) = \frac{e^{x_i}}{\sum_{j=1}^{J} e^{x_j}} i = 1, ..., J$
tanh	$f(x) = \tanh(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$

Table 1.1: Examples of activation functions, operating either element-wise or vector-wise, depending on the function

MSE / L2 Loss / Quadratic Loss	$\frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{N}$
(Binary) Cross Entropy (average reduction on higher dimensions)	$\frac{\sum_{i=1}^{N} \sum_{j=1}^{C} \hat{y}_i \log (y_{i,j})}{N}$
Categorical Cross Entropy (sum reduction on higher dimensions)	$-\sum_{i=1}^{N} \hat{y}_i + \log \left(\sum_{i=1}^{N} \sum_{j=1}^{C} y_{i,j} \right)$

Table 1.2: y is the output of the network, N is the batch size multiplied by the number of outputs (e.g. pixels), C is the number of classes and $\hat{y}$ is the correct output.

Chapter 2

SPICE model Validation

In order to properly analyze more critical components in terms of losses , is useful to first create a digital model of the device,a validation procedure of the SPICE models of each device has been performed. The following sections describes the procedure and results of this validation procedure.

2.1 Fast Leg GaN FET model validation

This section describes the behaviour of the SPICE model of the EPC2304,compared with EPC datasheet [2]

2.1.1 Static charateristics

Output I-V charateristics

```
.temp 25

.dc Vds 0 6 0.1 Vgs list 2 3 4 5
```

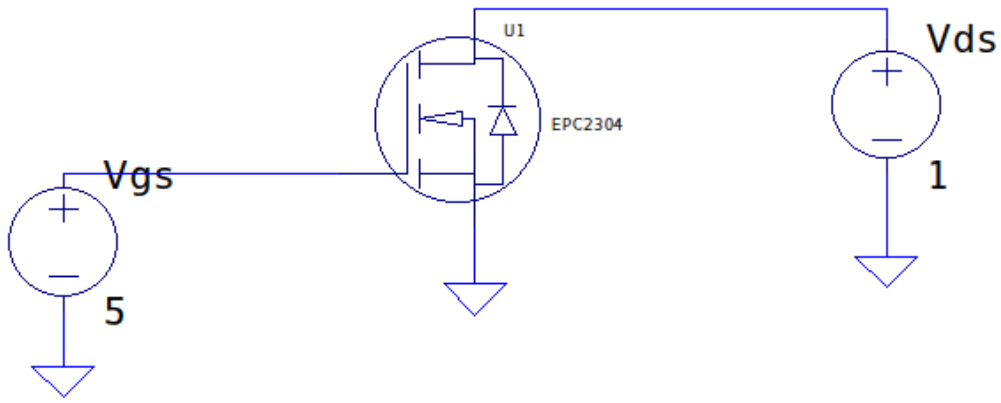

Figure 2.1: Schematic for I-V validation

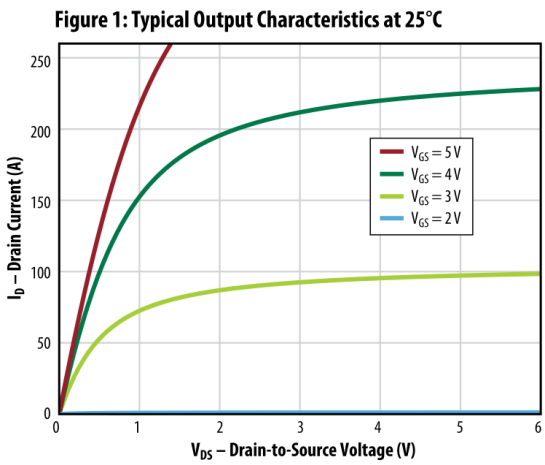

Figure 2.2: Datasheet I-V

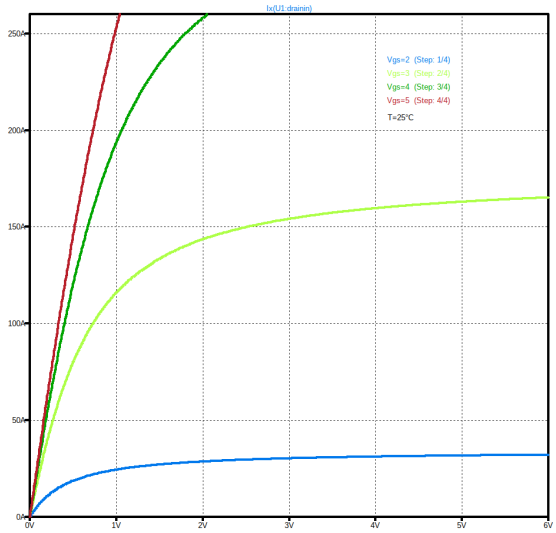

Figure 2.3: SPICE I-V

Vds@Vgs	Ids (datasheet)	Ids (simulation)
Vds=1V;Vgs=2V	$\approx 0A$	25.13A
Vds=6V;Vgs=3V	$\approx 100A$	163.15A
Vds=1V;Vgs=4V	$\approx 150A$	187.12A
Vds=1V;Vgs=5V	$\approx 215A$	253.07A

Table 2.1: I-V data comparision

Trans charateristics

```
.step temp 25 125 100

.temp 25

.dc Vgs 0.5 5 0.1
```

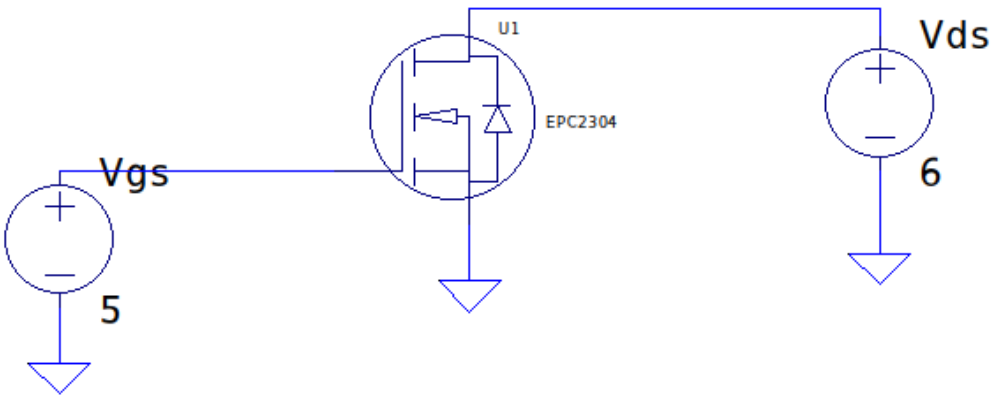

Figure 2.4: Schematic for trans charateristics validation

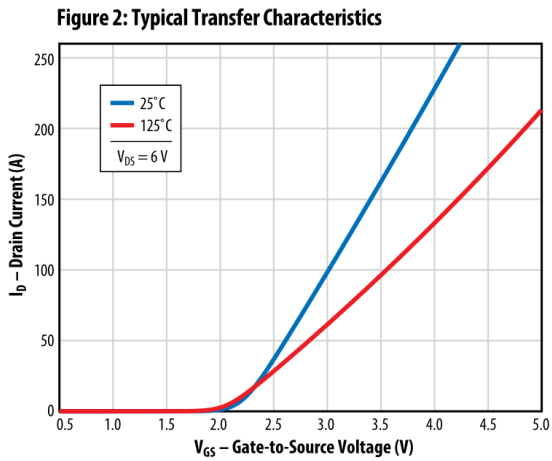

Figure 2.5: Datasheet trans charateristics

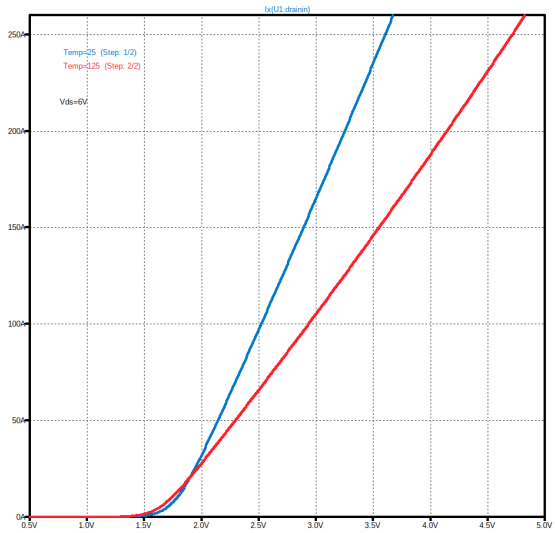

Figure 2.6: SPICE trans charateristics

Comparing the 2 it seems like the real device has a Vth of $\approx 2V$,while the simulated one has a Vth of $\approx 1.5V$

Vgs@T	Ids (datasheet)	Ids (simulation)
Vgs=2.5V;T=25	$\approx 40A$	98.58A
Vgs=2V;	$\approx 100A$	163.15A
Vgs=2.5;	$\approx 150A$	187.12A
Vgs=4V;	$\approx 215A$	253.07A
Vgs=3V;T=25	$\approx 100A$	160.47A

Table 2.2: I-V data comparision

Rdson

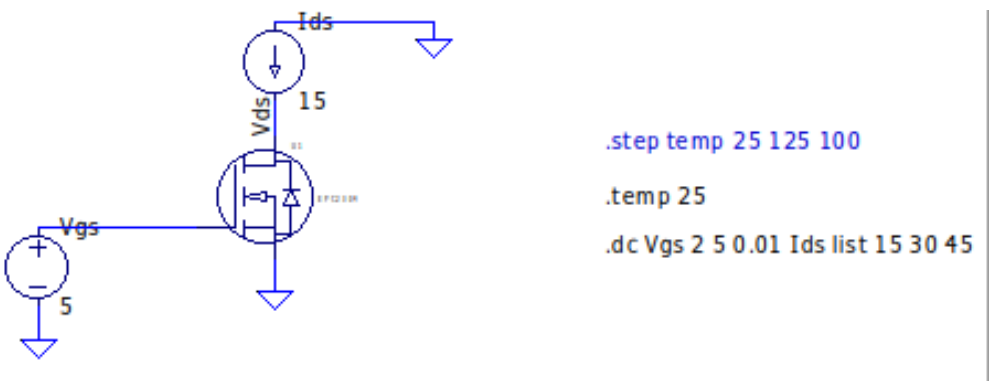

Figure 2.7: Schematic for Rdson validation

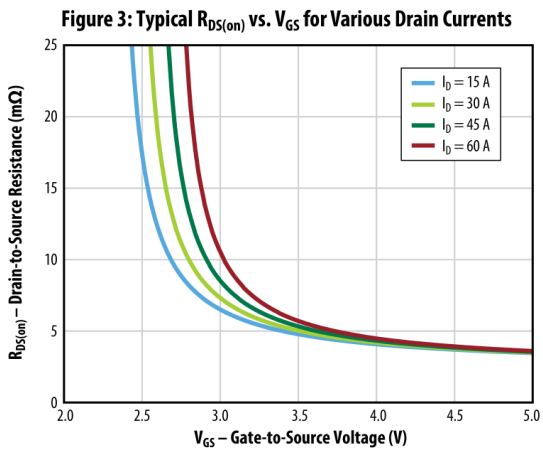

Figure 2.8: Datasheet Rdson

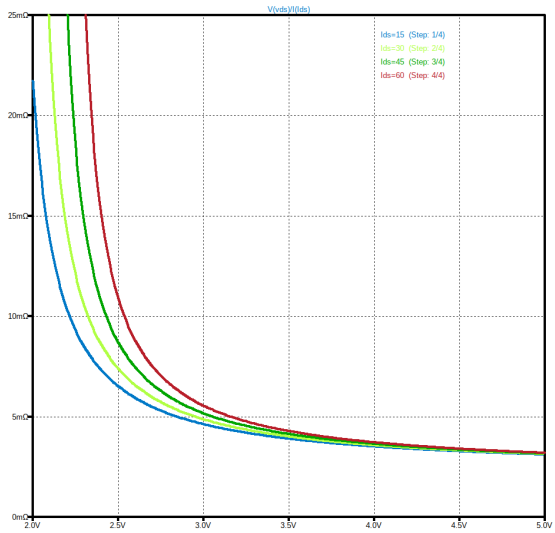

Figure 2.9: SPICE Rdson

Vgs@Ids	Rds (datasheet)	Rds (simulation)
Vgs=2V;Id=15A	$\approx 17.5m\Omega$	$6.8m\Omega$
Vgs=3V;Ids=60A	$\approx 10m\Omega$	6Ω
Vgs=2V;Ids=15A	$+\infty$	$22m\Omega$
Vgs=5V	$\approx 4m\Omega$	$3.78m\Omega$

Table 2.3: Rds data comparision

Rdson vs tempearture

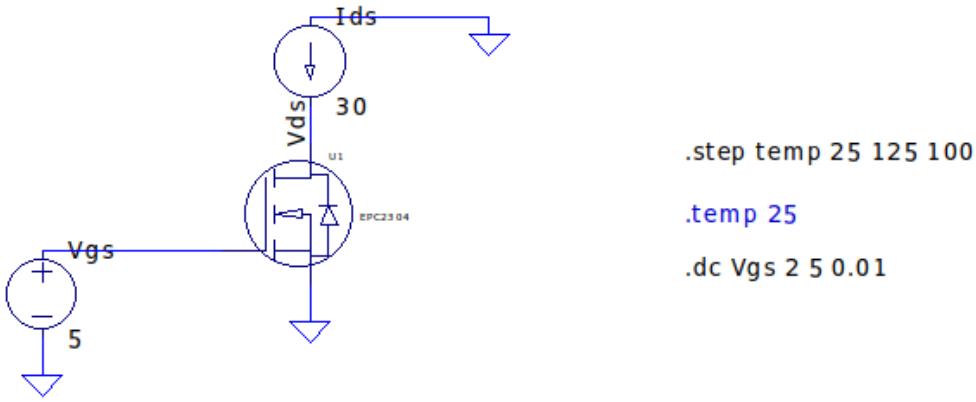

Figure 2.10: Schematic for Rdson vs tempearture

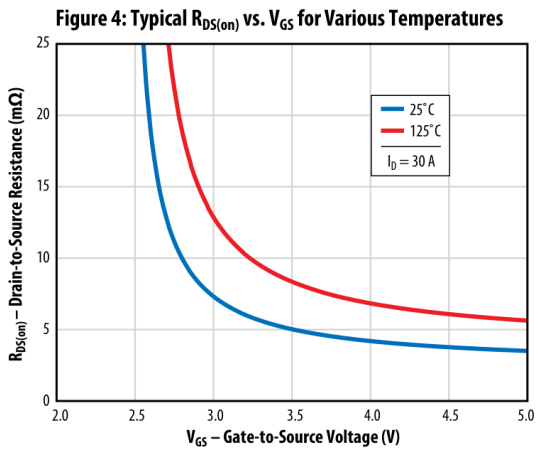

Figure 2.11: Datasheet Rdson vs tempearture

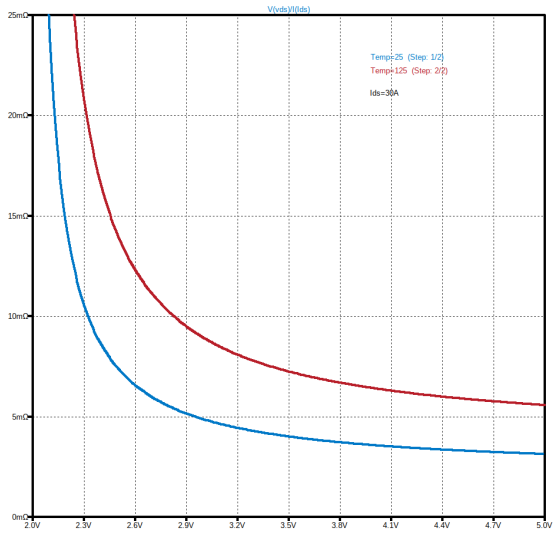

Figure 2.12: SPICE Rdson vs tempearture

Vgs@T	Rds (datasheet)	Rds (simulation)
Vgs=3.5V;T=25°C	$\approx 5m\Omega$	$4.6m\Omega$
Vgs=5V;T=125°C	$\approx 5.5m\Omega$	5.8Ω
Vgs=2.95V;T=25°C	$\approx 7m\Omega$	$5m\Omega$
Vgs=2.3V@T=25°C	$+\infty$	$10m\Omega$
Vgs=2.3V@T=125°C	$+\infty$	$21.2m\Omega$

Table 2.4: Rds temperature comparision

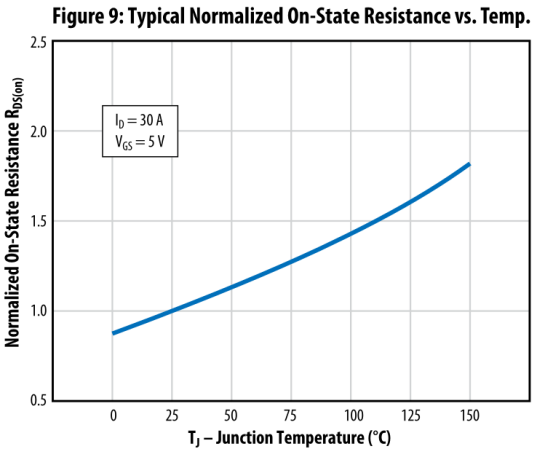

Figure 2.13: Datasheet Rdson vs temperature

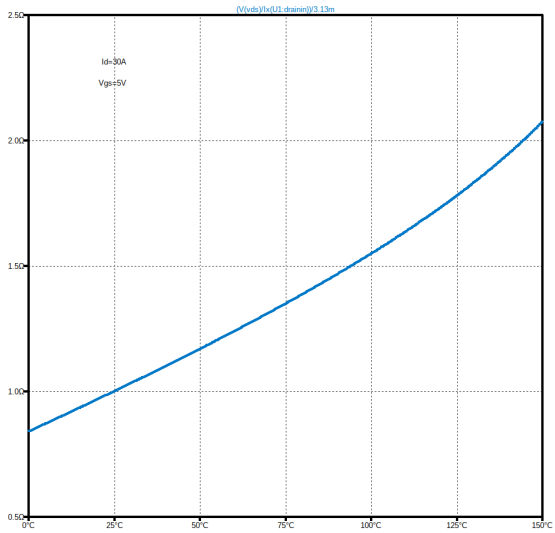

Figure 2.14: SPICE Rdson vs temperature

Reverse body diode

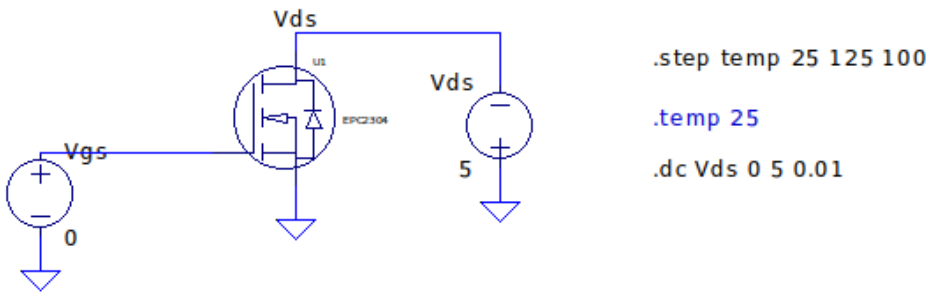

Figure 2.15: Schematic for Reverse body diode

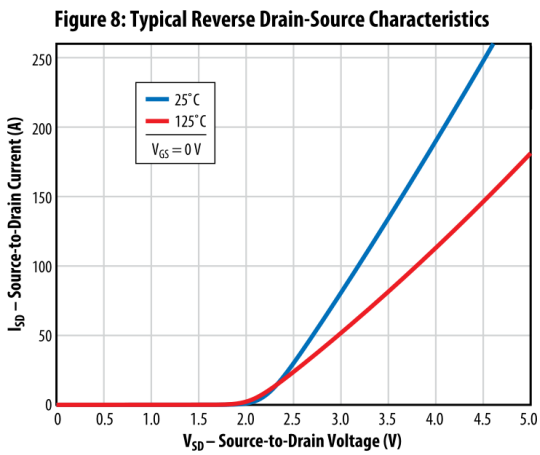

Figure 2.16: Datasheet Reverse body diode

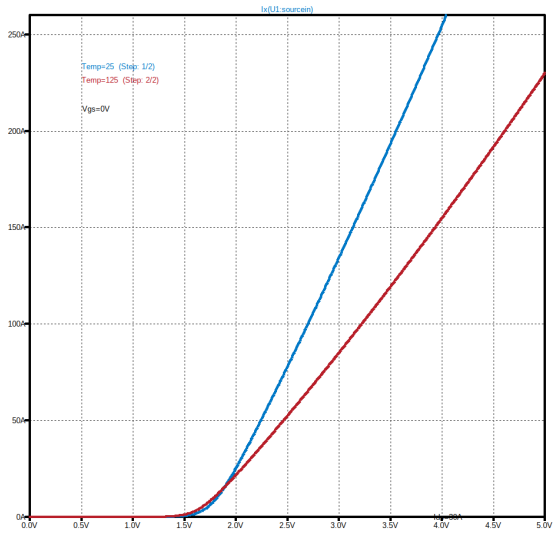

Figure 2.17: SPICE Reverse body diode

Vgs@T	Rds (datasheet)	Rds (simulation)
Vss=4.5V;T=25°C	$250A$	$317A$
Vsd=3V;T=125°C	$\approx 50A$	$88.7A$
Vsd=4.5V;T=125°C	$\approx 148A$	$191A$

Table 2.5: Reverse diode data comparision

Vth vs temperature

Figure 2.18: Schematic for Vth vs temperature

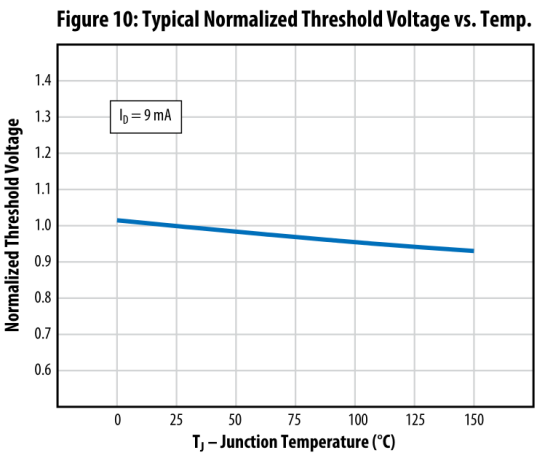

Figure 2.19: Datasheet Vth vs temperature

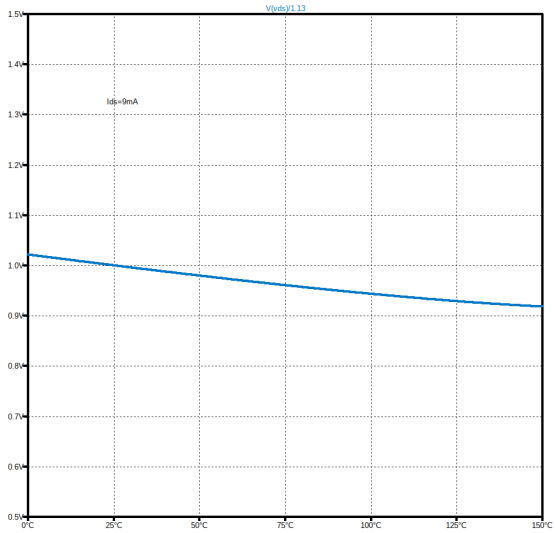

Figure 2.20: SPICE Vth vs temperature

2.1.2 Dynamic Charateristics

Typical switching curves

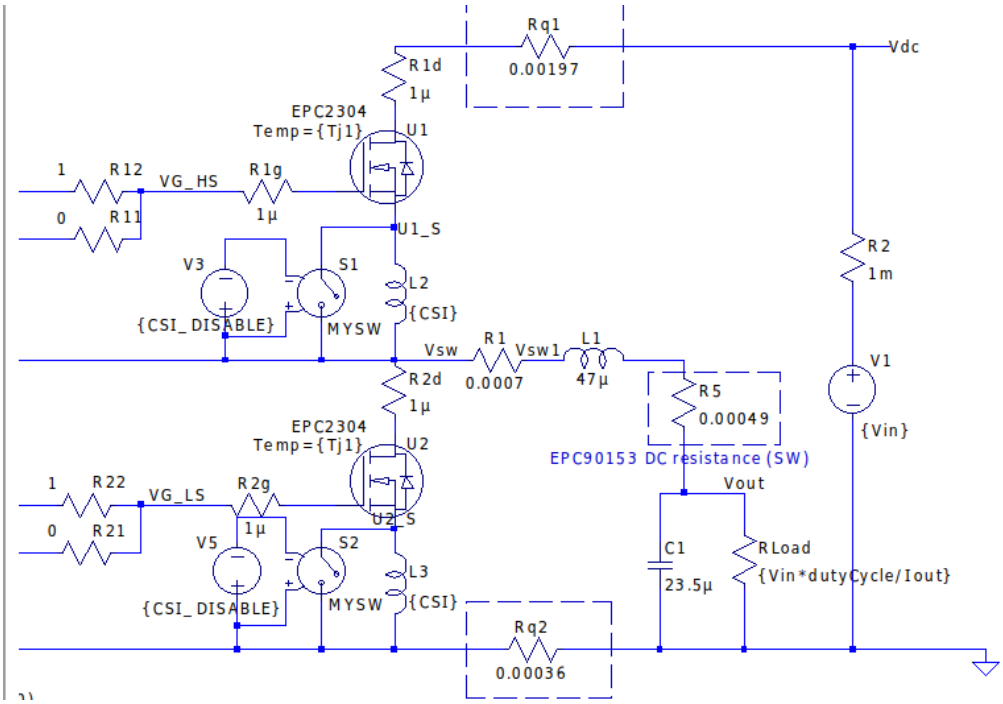

Figure 2.21: Schematic for switching behaviour

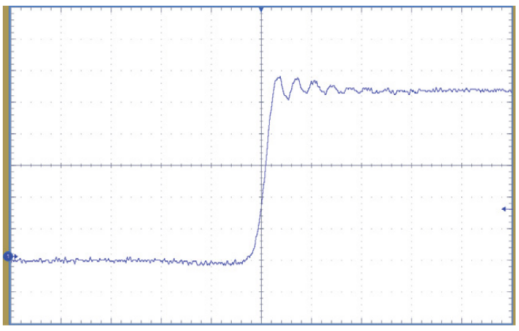

Figure 2.22: Datasheet Turn on curve

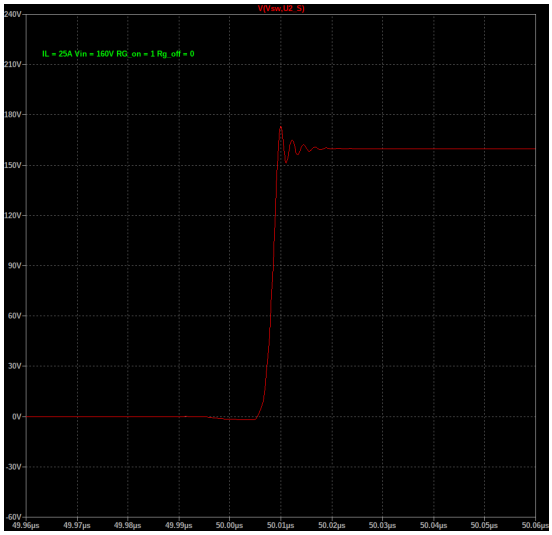

Figure 2.23: SPICE Turn on curve

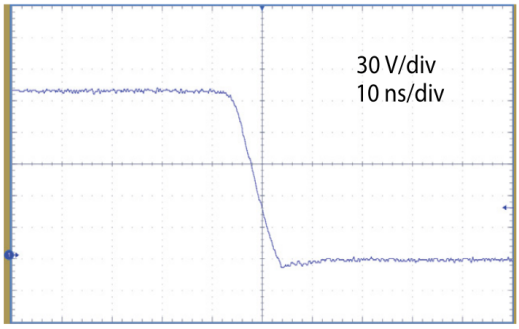

Figure 2.24: Datasheet Turn off curve

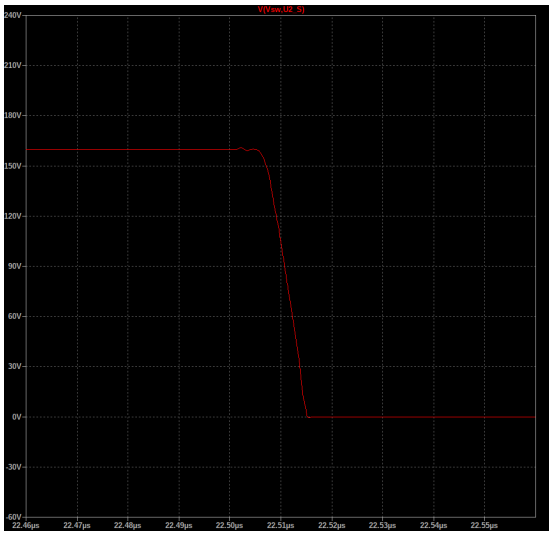

Figure 2.25: SPICE Turn off curve

Typical gate charge

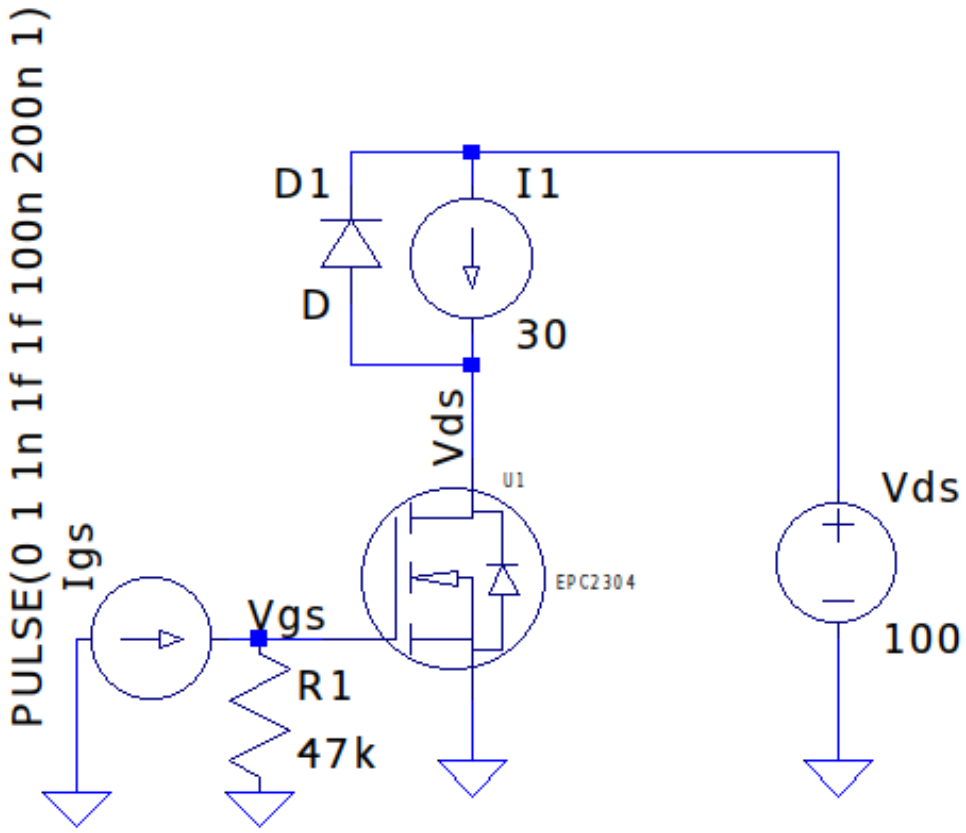

Figure 2.26: Schematic for gate charge

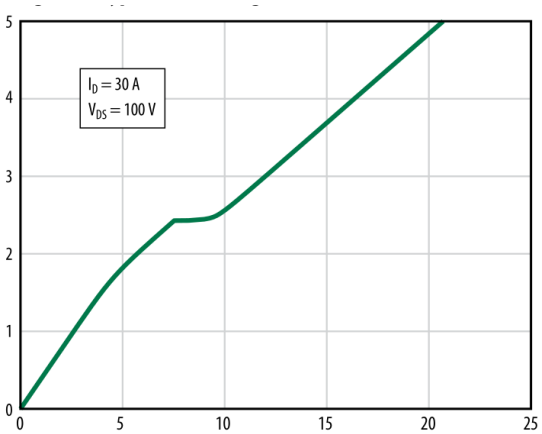

Figure 2.27: Datasheet gate charge

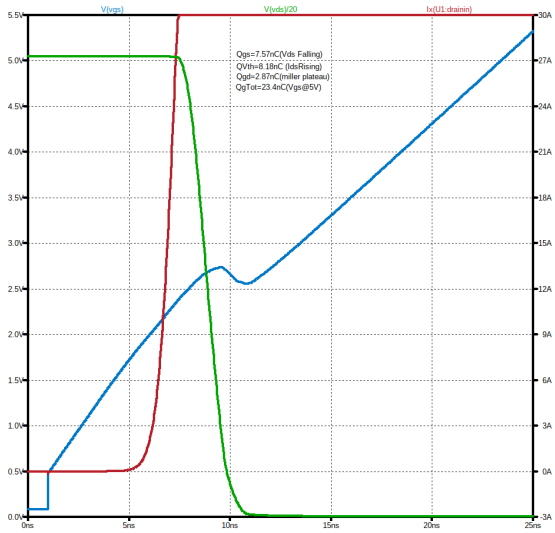

Figure 2.28: SPICE gate charge

Q (datasheet)	Q (simulation)
Qgtot=21nC	Qgtot=23.4nC
Qgs=7.5nC	Qgs=7.57nC
Qgd=2nC	Qgd=2.87nC
QVth=5.2nC	QVth=8.18nC

Table 2.6: Gate charge data comparision

2.2 Capacitors model validation

This section briefly describes the SPICE model behaviour compared to the datasheet described behaviour

Flying capacitor

no particular notes needed,model fits real behaviour,no temperature model available

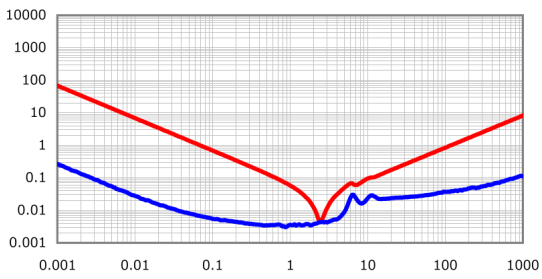

Figure 2.29: Datasheet Z

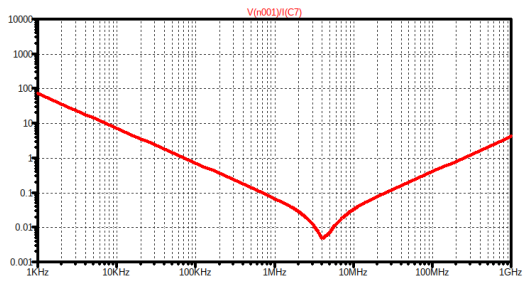

Figure 2.30: SPICE Z

HF decoupling capacitor

no particular notes needed,model fits real behaviour,no temperature model available

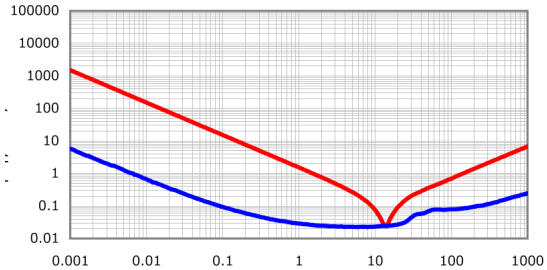

Figure 2.31: Datasheet Z

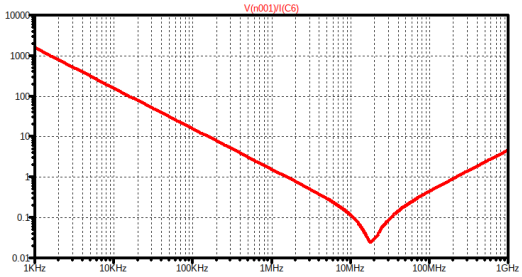

Figure 2.32: SPICE Z

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Bibliography

[1] S. Zhang, C. Zhu, J. K. O. Sin, and P. K. T. Mok. «A Novel Ultrathin Elevated Channel Low-temperature Poly-Si TFT». In: 20 (Nov. 1999), pp. 569–571 (cit. on p. 3).

[2] EPC-CO. *eGaN® FET DATASHEET EPC2304*. `https://epc-co.com/epc/Portals/0/epc/documents/datasheets/EPC2304_datasheet.pdf`. 2025 (cit. on p. 6).